

Zach Greenberg

Mechanical Engineering Student Specializing in Mechatronics & Advanced Prototyping

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Education

Georgia Institute of Technology, Atlanta, GA

Expected Graduation: *May 2027, May 2028*

- Candidate for Bachelor of Science in Mechanical Engineering
- Candidate for Masters of Science in Mechanical Engineering

GPA 3.84

GPA 4.00

Experience

Assistant Project Manager Intern at Batchelor & Kimball, Atlanta GA

May 2026 - Present

- Managed execution and mechanical coordination of industrial HVAC and cooling infrastructure for data centers
- Optimized cooling workflows and equipment layouts to maximize thermal efficiency in high density facilities
- Conducted system risk assessments and root cause analysis during complex piping and hardware installation
- Engineered a schedule tracking tool to identify critical path conflicts and accelerate equipment delivery

Automation and Controls Intern at Dennis Group, Salt Lake City, UT

May 2025 - August 2025

- Developed Python and Studio 5000 logic enabling integration of third-party hardware with Allen-Bradley PLCs
- Contributed to electrical and controls design, writing PLC logic, and creating animated HMI interfaces in Ignition
- Developed and animated machine screens, debugged automation issues, and supported system commissioning

Invention Studio Casting Master, Atlanta, GA – Awarded Best New PI of Spring 25

March 2025 - Present

- Oversaw foundry operations, tooling procurement, equipment maintenance, and user safety training programs
- Facilitated operations in nation's largest student-run makerspace, managing 3D printers and CNC machines
- Trained and mentored 100+ students in hands-on fabrication, guiding varied projects from concept to execution
- Maintained metalworking and woodworking equipment, ensuring strict safety standards and tool calibration

Internship at Mobility Prosthetics, Murray, UT

June 2023 - August 2023, May 2024 - July 2024

- Enhanced prosthetic fitting through refining and modeling 3D prints and managing timely print jobs
- Organized and managed a computer system to keep a record of placement, progress, and receipt of orders
- Created an organizational system for hardware to be sorted by patient name and type of equipment

Projects (see portfolio link for more detail)

Marine Robotics, Georgia Tech, Atlanta, GA

August 2023 - Present

RoboSub - the creation of an autonomous underwater vehicle tasked with completing several challenges

- Engineered electromechanical manipulation and collection systems for an autonomous underwater vehicle
- Conducted pressure testing and FEA to ensure the structural integrity of sealed housings at operational depths
- Fabricated custom electronics housings and thermal management structures to protect critical internal systems

Medical Robotics, Georgia Tech, Atlanta GA

August 2023 - December 2024

Humanoid Robotics - Spearheaded creation of a humanoid robotic hand for medical research

- Optimized finger kinematics and thumb opposition with rapid FDM/SLA prototyping and mechanical validation
- Integrated sub-miniature actuation systems within strict anatomical space constraints to ensure fluid motion
- Transitioned the final prototype to clinical testing phases, where it is currently undergoing amputee evaluation

Research

EPIC Labs, Georgia Tech | Mechanical Lead, Wearable Motion Capture Research

January 2026 - Present

- Spearheaded mechanical design of wireless motion capture suits for prosthetic and exoskeleton testing
- Engineered ergonomic sensor enclosures and fabric closures utilizing SLA/FDM rapid prototyping
- Defined mechanical constraints to maintain sensor orientation and data integrity during dynamic movement
- Documented hardware iterations and validation protocols to prepare the platform for clinical subject trials

University of Utah, Neurosurgery Research Assistant, Salt Lake City, UT

June 2023 - August 2023

MEMBRANE Study: Endovascular Embolization Device Clinical Trial (TRUFILL®)

- Collected and recorded performance data in capillary structures to improve recovery and limit recurrence
- Analyzed and organized study data, procedures, and results from patient records and study booklets
- Streamlined the study's processes through the structuring and categorization of research files

Skills

Mechanical Design: DFM, Kinematics, FEA (Static/Thermal), GD&T, Mechanism Design, Thermal Management

CAD & Fabrication: SolidWorks, Fusion 360, 3D Printing (SLA/FDM), CNC Machining, Metalworking, Woodworking

System & Controls: Python, C++, MATLAB, Studio 5000 (PLC), Ignition (HMI), Arduino, Simulink, GIT/GitHub